ZHANG Bin, Ph.D. Candidate

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Education

Sep. 2021 – Now	**Ph.D. Student** in *Mechanical Engineering*, **The Hong Kong Polytechnic University**.
Mar. 2024 – Jul. 2024	**Visiting Ph.D. Student, KTH Royal Institute of Technology**.
Sep. 2017 – Jul. 2020	**M.Sc.** in *Control Science and Engineering*, **China Academy of Space Technology**.
Sep. 2013 – Jul. 2017	**B.Eng.** in *Detection, Guidance, and Control Technology*, **Beihang University**.

Employment History

May 2021 – Aug. 2021	**Research Assistant** at ROMI Lab, The Hong Kong Polytechnic University.
Aug. 2020 – Mar. 2021	**Research Intern** at Qian Xuesen Laboratory, China Academy of Space Technology.

Research Interests

- Multi-agent formation control.
- Multi-robot manipulation.
- Control theory and applications.

Research Publications

Equal contribution

Journal Articles

1. **Xiaodong Shao**#, **Bin Zhang**#, H. Zhi, J. G. Romero, B. Fan, Q. Hu, and D. Navarro-Alarcon, "InPTC: Integrated planning and tube-following control for prescribed-time collision-free navigation of wheeled mobile robots," *IEEE/ASME Transactions on Mechatronics*, vol. 30, no. 5, pp. 3339–3350, Oct. 2025. doi: 10.1109/TMECH.2024.3486727

2. **Bin Zhang**, H. Zhi, J. G. Romero, and D. Navarro-Alarcon, "Leader-follower formation control of perturbed nonholonomic agents along parametric curves with directed communication," *IEEE Robotics and Automation Letters*, vol. 9, no. 10, pp. 8603–8610, Aug. 2024. doi: 10.1109/LRA.2024.3445657

3. **Bin Zhang**, H. Zhi, J. G. Romero, L. Labazanova, A. Duan, X. Li, and D. Navarro-Alarcon, "Fourier-based multi-agent formation control to track evolving closed boundaries," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 70, no. 11, pp. 4549–4559, Nov. 2023. doi: 10.1109/TCSI.2023.3300772

4. **Bin Zhang** and J. Zhou, "Characteristic model-based station-keeping control for halo orbit," *Acta Aeronautica et Astronautica Sinica*, vol. 40, no. 11, Nov. 2019. (in Chinese) doi: 10.7527/S1000-6893.2019.23306

5. Y. Tian, **Bin Zhang**, X. Shao, and D. Navarro-Alarcon, "Encircling general 2-d boundaries by mobile robots with collision avoidance: A vector field guided approach," *IEEE/ASME Transactions on Mechatronics*, Early Access 2026. doi: 10.1109/TMECH.2026.3663573

6. N. Han, G. Gong, **Bin Zhang**, Y. Xu, B. Yang, Y. Liu, and D. Navarro-Alarcon, "Prescribed performance control of deformable object manipulation in spatial latent space," *IEEE/ASME Transactions on Mechatronics*, Early Access 2026. doi: 10.1109/TMECH.2026.3657722

7. H. Zhi, **Bin Zhang**, J. Qi, J. G. Romero, X. Shao, C. Yang, and D. Navarro-Alarcon, "Non-prehensile object transport by nonholonomic robots connected by linear deformable elements," *IEEE Robotics and Automation Letters*, vol. 9, no. 10, pp. 8651–8658, Aug. 2024. doi: 10.1109/LRA.2024.3440096

8. M. Muddassir, G. Limbert, **Bin Zhang**, A. Duan, J.-J. Tan, and D. Navarro-Alarcon, "Model predictive thermal dose control of a robotic laser system to automate skin photorejuvenation," *IEEE/ASME Transactions on Mechatronics*, vol. 28, no. 2, pp. 737–747, Apr. 2023. doi: 10.1109/TMECH.2022.3218806

9. W. Ma, **Bin Zhang**, L. Han, S. Huo, H. Wang, and D. Navarro-Alarcon, "Action planning for packing long linear elastic objects into compact boxes with bimanual robotic manipulation," *IEEE/ASME Transactions on Mechatronics*, vol. 28, no. 3, pp. 1718–1729, Jun. 2023. doi: 10.1109/TMECH.2022.3223990

10. H. Y. Lee, P. Zhou, **Bin Zhang**, L. Qiu, B. Fan, A. Duan, J. Tang, T. L. Lam, and D. Navarro-Alarcon, "A distributed dynamic framework to allocate collaborative tasks based on capability matching in heterogeneous multi-robot systems," *IEEE Transactions on Cognitive and Developmental Systems*, vol. 16, no. 1, pp. 251–265, Feb. 2023. doi: 10.1109/TCDS.2023.3264034

11. S. Huo, **Bin Zhang**, M. Muddassir, D. T. W. Chik, and D. Navarro-Alarcon, "A sensor-based robotic line scan system with adaptive ROI for inspection of defects over convex free-form specular surfaces," *IEEE Sensors Journal*, vol. 22, no. 3, pp. 2782–2792, Feb. 2022. doi: 10.1109/JSEN.2021.3132428

12. J. Zhou, J. Hu, Y. Bai, and **Bin Zhang**, "Optimal impulsive time-fixed transfers around the libration points of the restricted three-body problem," *Astrophysics and Space Science*, vol. 365, pp. 1–18, May 2020. doi: 10.1007/s10509-020-03793-7

13. J. Zhou, J. Hu, and **Bin Zhang**, "Approximate analytical solutions of motion near the collinear libration-points in restricted three-body problem," *Journal of Deep Space Exploration*, vol. 7, no. 1, Feb. 2020. (in Chinese) doi: 10.15982/j.issn.2095-7777.2020.20190408001

14. J. Zhou, J. Hu, and **Bin Zhang**, "Analytical solutions for relative motion in the cicular restricted three-body problem," *Journal of Astronautics*, vol. 41, no. 2, Feb. 2020. (in Chinese) doi: 10.3873/j.issn.1000-1328.2020.02.004

Conference Proceedings

1. **Bin Zhang**, C. Li, and J. Zhou, "Station-keeping of halo orbits using convex optimization-based receding horizon control," in *2021 China Automation Congress (CAC)*, Oct. 2021, pp. 7160–7165. doi: 10.1109/CAC53003.2021.9728345

2. **Bin Zhang** and J. Zhou, "Characteristic model-based station-keeping control for halo orbits about Sun-Earth L1 point," in *2019 Chinese Automation Congress (CAC)*, Nov. 2019, pp. 418–423. doi: 10.1109/CAC48633.2019.8997351

3. A. Duan, **Bin Zhang**, L. Hu, L. Feng, and D. Navarro-Alarcon, "Balancing control of wheeled-legged hexapedal robots for traversing rough terrains," in *2022 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, Dec. 2022, pp. 1905–1910. doi: 10.1109/ROBIO55434.2022.10011895

Skills

Languages	English, Chinese (*mother tongue*).
Coding	Python, C/C++, Matlab, HTML, LaTeX.
Others	ROS, Linux, Webots, Arduino, SolidWorks.

Miscellaneous Experience

Awards and Achievements

2017	**Outstanding Graduate of Beihang University.**
2015	**National Inspirational Scholarship.**
2014	**National Inspirational Scholarship.**

Academic Services

Reviewer	ICRA 2022, 2024-2026; CASE 2024.
	IEEE Control Systems Letters.
	Journal of Field Robotics.
	IEEE Robotics and Automation Magazine.
	IEEE Transactions on Control of Network Systems.
	Aerospace Science and Technology.
	IEEE Transactions on Robotics.
	IEEE Robotics and Automation Letters.

Teaching Experience

2021	**Teaching Assistant**, Perceptual Robotics.
2022	**Teaching Assistant**, Linear Systems and Control.
	Teaching Assistant, Advanced Control Technology.
2023	**Teaching Assistant**, Linear Systems and Control.